

Haramaya University
College of Computing and Informatics
Department of Computer Science

Industrial Practice Report Guideline (Software Development)

Important Notes:

- ✓ Weekly contact with your advisor is **mandatory**.
 - ✓ Strict adherence to this guideline is required.
 - ✓ Report any issues to the department promptly.
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Report Structure

Preliminary Sections

1. **Front Page**
 2. **Cover Page**
 3. **Acknowledgement**
 4. **List of Abbreviations**
 5. **Table of Contents**
 6. **List of Tables/Figures** (if applicable)
 7. **Organization of the Document**
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Chapter 1: Introduction

1. **Background of the Organization**
 - Overview of the host organization.
2. **Structure of the Organization**
 - Hierarchy and relevance to the project.

3. Team/Department Overview

- Roles, workflow, and collaboration methods.

4. Project Context

- Purpose, goals, scope, and problem statement.

5. Objectives

- **General:** Broad aim of the project.
 - **Specific:** Measurable outcomes.
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Chapter 2: Software Development Plan

1. Project Schedule

- Milestones, deliverables, and timelines (Gantt charts recommended).

2. Resources

- Tools (software/hardware), workspaces, and team roles.

3. Risk Management

- Risk analysis, mitigation strategies, and contingency plans.

4. Development Methodology

- Agile (Scrum/Kanban), Waterfall, or hybrid; justify choice.

5. Quality Assurance

- Testing protocols, code reviews, and documentation standards.

6. Required Documents

- Technical specs, user manuals, and design documents.
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Chapter 3: System Requirements

1. Existing System Analysis

- Overview and limitations.

2. Proposed System

- **Functional Requirements:** Features and capabilities.
 - **Non-Functional Requirements:**
 - Usability, reliability, performance, security, legal compliance.
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Chapter 4: System Analysis & Design

1. System Models

- Use case diagrams, class/object diagrams, sequence/activity diagrams.

2. User Interface Design

- Mock-ups and navigational flow.

3. Software Architecture

- Subsystem decomposition, deployment diagrams, data management.

4. Security & Access Control

- Authentication, authorization, and encryption methods.
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Chapter 5: Implementation

1. Development Process

- Tools used (e.g., IDE, frameworks, version control).

2. Challenges & Solutions

- Technical hurdles and how they were resolved.

3. Testing

- Unit, integration, and system testing results.
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Chapter 6: Internship Reflection & Conclusion

1. Learning Outcomes

- Practical skills, theoretical knowledge, and personal growth.

2. Challenges Faced

- Workplace adaptability, technical issues, etc.

3. Recommendations

- For the organization and future internship programs.
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Appendices

- Code snippets, full diagrams, survey questionnaires (if applicable).

References

- Cite all sources using IEEE/APA format.